

WEEK 8

# Week 8: Advanced Storage Concepts - Lab Guide

HPC & Cluster Computing

# Week 8 Lab Part 2: Deploying BeeGFS

**Objective**: Deploy a high-performance Parallel File System. **Goal**: Create a shared `/scratch` space using the local disks of the Compute Nodes.

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## 1. Prerequisites

- [ ] **Master Node**: Will host the `mgmtd` and `meta` services.
- [ ] **Compute Nodes**: Will host the `storage` service (Converged Architecture).
- [ ] **Networking**: BeeGFS relies heavily on DNS. Ensure `/etc/hosts` works on all nodes.

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## 2. Install BeeGFS Repositories

We need to add the BeeGFS repo to all nodes.

- [ ] Get the repo file: `bash sudo wget -O /etc/yum.repos.d/beegfs-rhel9.repo https://www.beegfs.io/release/beegfs_7_4/dists/beegfs-rhel9.repo`

### 2.2 On Compute Nodes (via Warewulf)

Add the repo to your node image/container (e.g., `rocky-9`):

- [ ] Enter the container:

`bash sudo wwctl container exec rocky-9 /bin/bash`

- [ ] Add the repo (Inside Container):

`bash wget -O /etc/yum.repos.d/beegfs-rhel9.repo https://www.beegfs.io/release/beegfs_7_4/dists/beegfs-rhel9.repo`

- [ ] Exit and Rebuild:

`bash exit sudo wwctl container build rocky-9`

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## 3. Configure Master Services (Mgmtd & Meta)

### 3.1 Install

- [ ] Install packages: `bash sudo dnf install beegfs-mgmtd beegfs-meta beegfs-utils -y`

### 3.2 Initialize Management Service

- [ ] Setup Mgmtd service: `bash sudo /opt/beegfs/sbin/beegfs-setup-mgmtd -p /var/beegfs/mgmtd -S beegfs-mgmtd`

- [ ] Setup Meta service: `bash sudo /opt/beegfs/sbin/beegfs-setup-meta -p /var/beegfs/meta -s 2 -m master` *(Explanation: `-s 2` sets the ServiceID, `-m master` points to the Management server).*

### 3.4 Start Master Services

- [ ] Enable and Start: `bash sudo systemctl enable --now beegfs-mgmtd beegfs-meta`

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## 4. Configure Compute Services (Storage)

Now we turn our Compute Nodes into Storage servers.

### 4.1 Install (Inside Container)

The storage service binary must exist in the image.

- [ ] Install storage package:

```
bash sudo wwctl container exec rocky-9 dnf install beegfs-storage -y
```

- [ ] Rebuild and Reboot:

```
bash sudo wwctl container build rocky-9 sudo wwctl power cycle node[01-04]
```

### 4.2 Initialize Storage Service (On Each Node)

Log into `node01` (and similarly for others):

- [ ] Create local storage directory:

```
bash mkdir -p /data/beegfs/storage
```

```
bash # Example for Node01 (ID 301) sudo /opt/beegfs/sbin/beegfs-setup-storage -p /data/beegfs/storage -s 3 -i 301 -m master
```

- [ ] Start the service:

```
bash sudo systemctl enable --now beegfs-storage
```

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## 5. Mount the Filesystem (Client)

Now that the servers are running, let's mount the filesystem.

### 5.1 Install Client

- [ ] Install client packages: `bash sudo dnf install beegfs-client beegfs-helperd beegfs-utils -y`

### 5.2 Configure Client

- [ ] Point client to Master: `bash sudo /opt/beegfs/sbin/beegfs-setup-client -m master`

### 5.3 Start Client

- [ ] Start client service: `bash sudo systemctl enable --now beegfs-helperd beegfs-client`
- [ ] **Verification** Check if mounted: `bash df -h | grep beegfs # Should see: beegfs_nodev ... /mnt/beegfs`

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## 6. Verification & Striping Test (Interactive)

### 6.1 Check Health

On the Master, check connectivity:

- [ ] List Metadata Nodes:

```
bash beegfs-ctl --listnodes --nodetype=meta
```

- [ ] List Storage Nodes:

```
bash beegfs-ctl --listnodes --nodetype=storage
```

*Result*: Do you see all your compute nodes listed?

### 6.2 Stripe Test

Create a file, write to it, and see where the chunks go.

- [ ] Set stripe pattern (Spread across 2 nodes):

```
bash beegfs-ctl --setpattern --numtargets=2 /mnt/beegfs
```

- [ ] Create a 1GB file:

```
bash dd if=/dev/zero of=/mnt/beegfs/testfile bs=1M count=1000
```

- [ ] Check chunk distribution:

```
bash beegfs-ctl --getentryinfo /mnt/beegfs/testfile
```

*Observation*: You should see that the file is split across, for example, Target ID 301 (Node 1) and Target ID 401 (Node 2).

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## 7. Submission



- [ ] `beegfs-ctl --listnodes --nodetype=storage` showing all your nodes.
- [ ] `df -h` showing the `/mnt/beegfs` mount.